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Detailed Measurements on a Modern Combustor Dump Diffuser System

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An experimental investigation has been carried out to determine the flow characteristics and aerodynamic performance of a modern gas turbine combustor dump diffuser. The system comprised a straight walled prediffuser, of area ratio 1.35, which projected into a dump cavity where the flow divided to pass either into the flame tube or surrounding feed annuli. In addition, a limited amount of air was removed to simulate flow used for turbine cooling. The flame tube was relatively deep, having a radial depth 5.5 times that of the passage height at prediffuser inlet, and incorporated burner feed arms, cowl head porosity, cooling rings, and primary ports. Representative inlet conditions to the diffuser system were generated by a single-stage axial flow compressor. Results are presented for the datum configuration, and for a further three geometries in which the distance between prediffuser exit and the head of the flame tube (i.e., dump gap) was reduced. Relatively high values of stagnation pressure loss were indicated, with further significant increases occurring at smaller dump gaps. These high losses, which suggest a correlation with other published data, are due to the relatively deep flame tube and short diffuser length. Furthermore, the results also focus attention on how the presence of a small degree of diffuser inlet swirl, typical of that which may be found within a gas turbine engine, can result in large swirl angles being generated farther downstream around the flame tube. This is particularly true for flow passing to the inner annulus.

Introduction

The primary function of a gas turbine diffuser system is to decelerate air issuing from the compressor and to distribute this flow in a uniform and stable manner around the flame tube. This also needs to be achieved with a minimal loss of stagnation pressure due to the effect on specific fuel consumption and the penetration of flow entering the flame tube. In current advanced core engines these objectives are usually met with dump diffuser systems, which incorporate a prediffuser followed by a recirculating dump region surrounding the flame tube head. Although such systems have relatively high losses, they are favored because of their inherent flow stability and insensitivity to manufacturing tolerances and thermal expansions.

Fishenden and Stevens [1] initially investigated the overall performance of a dump system by varying several parameters, including the mass flow split to each feed annuli, prediffuser area ratio, and the distance between prediffuser exit and the flame tube head (i.e., dump gap D/h_2). All tests were conducted with a flame tube that had a radial depth, expressed in terms of the passage height at prediffuser inlet (W/h_1) of 3.5. It was concluded that the principal determinants of stagnation pressure loss in such systems are the amount of diffusion being attempted and the radius of curvature undertaken by the flow as it passes around the flame tube. This latter effect is a function of the size and shape of the flame tube and the dump gap. With this in mind it is interesting to note the more recent works of several authors [2–6], which illustrate the trend for flame tubes of increasing depth, with values (W/h_1) in excess of 4.0. In addition, this trend is likely to continue as stricter emission regulations are introduced. For aircraft applications there is also a constant desire to reduce weight, resulting in the demand for shorter length diffusers. These trends imply that, in a modern dump diffuser system, relatively large amounts of flow turning and curvature are required in the vicinity of the flame tube head, which has a significant effect on diffuser performance. It should also be noted that investigations by several authors have shown how the performance of any diffuser is affected by the flow conditions at inlet to the system. For example, Stevens et al. [7], Stevens and Wray [8], and Klein [9] noted the effects on diffuser performance due to wakes from upstream compressor outlet guide vanes. In addition, Lohmann et al. [10] considered tangential momentum within a prediffuser and showed that inlet swirl angles can increase as the flow is diffused due to reductions in axial velocity and flow radius. This latter effect may be of significance when considering the large changes in radius that must be undertaken by flow when passing around the head of the flame tube. It is therefore thought that as performance becomes more critical and designs less conservative, so it is necessary when investigating such systems to simulate accurately the inlet conditions that are typically generated by an upstream compressor.

This paper presents detailed measurements of the mean flow field and pressure distribution within a dump diffuser incorporating a flame tube of radial depth 5.5 times the passage height at pre-diffuser inlet (W/h_1). With the exception of Stevens et al. [2], this represents a significant increase in depth, and hence flow curvature, relative to other published data. In addition to the datum configuration, results are also presented for three other dump gaps. Thus, an assessment can be made of the performance penalty that will be incurred in such a system when the diffuser length is shortened by reducing the dump gap. All measurements were obtained on a fully annular facility with the diffuser located downstream of a single-stage axial flow compressor. This gave a good representation of the secondary flows, blade wakes, turbulence characteristics, rotor tip leakage, and other effects that are present at the inlet of a diffuser system operating within a gas turbine engine.

Experimental Facility

A comprehensive description of the experimental facility is given by Wray et al. [11]. Air is drawn from atmosphere into

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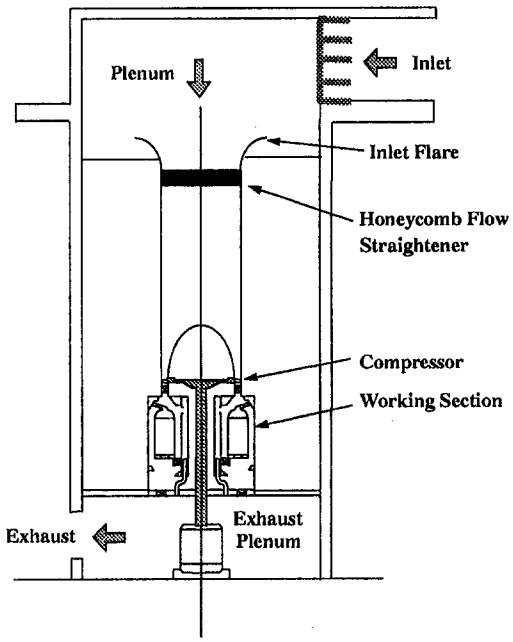
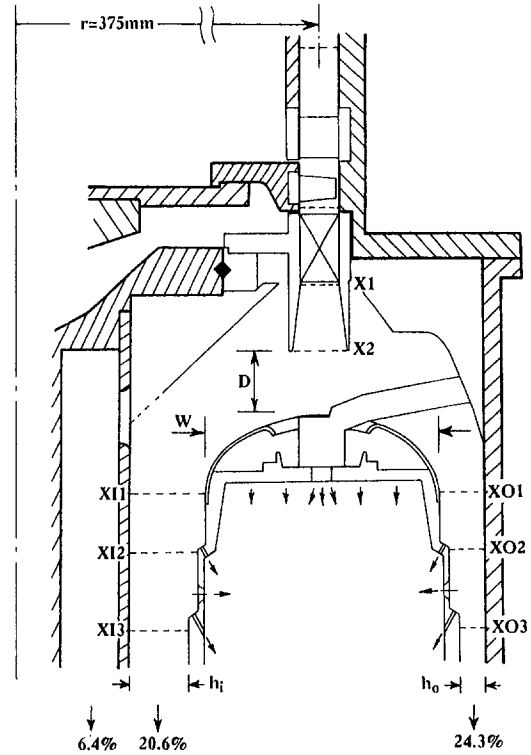


Fig. 1 Experimental facility



Pre-diffuser inlet height (h_1)	=	36.6mm
Pre-diffuser exit height (h_2/h_1)	=	1.35
Inner annulus height (h_i/h_1)	=	1.41
Outer annulus height (h_o/h_1)	=	0.67
Flame tube radial depth	=	5.5

Fig. 2 Diffuser geometry

a large plenum, above the vertically mounted facility, prior to passing through the entry flare, which also contains a honeycomb flow straightener (Fig. 1). The pressure rise required to draw air through the facility is provided by the single-stage axial flow compressor located immediately upstream of the working section. It consists of stationary inlet and outlet guide vane rows and a rotor, this being driven by a 45 k W DC motor sited in the exhaust plenum below the test rig. The rotor is operated at a constant nondimensional speed ($N/\sqrt{T}$) of 168.7, which, at a typical ambient temperature of 291 K, results in a rotor speed of 2880 rpm. This provides approximately 46 deg of inlet swirl to the compressor outlet guide vanes (OGVs), which are fully shrouded and representative of current engine design practice. Their operational Reynolds number of 2.6×10^5, based on blade true chord, together with the turbulence effects associated with the upstream rotor, means the blades are well above their critical transition value.

The inlet of the combustor diffuser system is located at the trailing edge of the OGVs where the passage height (h_1) is 36.6 mm with a radius of 375 mm at midpassage height (Fig. 2). For a mass flow of 4.6 kg/s, this gives a mean velocity at inlet to the prediffuser of approximately 45 m/s (Mach No. ≈ 0.13). The prediffuser is of area ratio 1.35 and has an axial length 1.5 times the inlet height (h_1). At prediffuser exit the flow enters the dump cavity, where it divides to pass into either the outer and inner feed annuli or flame tube. In addition, the results of Stevens et al. [2] have shown that removal of turbine cooling air from the dump region can significantly affect the flow properties within the feed annuli. Consequently a limited amount of air is removed through a series of plunged holes (ϕ43.5 mm) within the inner dump casing.

In its datum configuration, the distance between the exit of the prediffuser and the flame tube cowl corresponds to the annulus height at prediffuser exit (i.e., dump gap, $D/h_2 = 1.0$). Since the diffuser system is not canted, additional dump gaps of 0.80, 0.65, and 0.50 were obtained merely by raising the

Nomenclature

A = area
C_p = static pressure recovery coefficient
D = dump gas distance (Fig. 2)
h = annulus passage height
L = diffuser axial length (Fig. 14)
m = mass flow
N = compressor rotor speed
OGV = compressor outlet guide vane
p = local static pressure
P = local stagnation pressure
r = radius relative to rig centreline
r_i = inner casing radius relative to rig centerline
r_o = outer casing radius relative to rig centerline
T = ambient temperature, K
U = axial velocity
U_t = tangential velocity
W = flame tube radial depth
W' = radial flow deflection (Fig. 14)
λ = stagnation pressure loss coefficient
ρ = fluid density
θ = flow swirl angle

Superscripts

$\bar{}$ = area-weighted spatial mean value
$\tilde{}$ = mass-weighted spatial mean value
c = circumferentially weighted mean value

Subscripts

1, 2 = stations at prediffuser inlet, exit (Fig. 2)
$O1$, $O2$, $O3$ = stations within outer annulus (Fig. 2)
$I1$, $I2$, $I3$ = stations within inner annulus (Fig. 2)

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complete flame tube unit by the required distance. Porosity through the cowl head is provided by holes (ϕ79.5 mm) in each of the 20 burner sectors, this flow then entering the combustor primary zone through the cowl backplate. Also located within the center of each cowl hole is a burner, with representative porosity, and an attached fuel feed arm. Flow can also enter the flame tube through primary cooling rings, primary ports, and secondary cooling rings located in both the outer and inner feed annuli. At the entry plane to each annulus the radial depth of the flame tube is 5.5 times the passage height at prediffuser inlet; however, this ratio increases to 6.3 downstream of the second cooling ring. A series of throttles located below the working section allowed, for a given geometry, the correct mass flow split and overall mass flow to be set. All results presented in this paper were obtained for a mass flow split of:

Outer feed annulus	24.3%
Inner feed annulus	20.6%
Turbine bleed passage	6.4%
(Flame tube)	48.7%

[N.B. Food annulus and flame tube mass flows measured downstream of second cooling rings].

The division of mass flow to the turbine bleed passage and feed annuli were maintained to within ±0.1 and ±0.5 percent of their respective nominal values. The flow, having passed through the working section and throttle system, was expelled to atmosphere via the exhaust plenum below the facility.

Instrumentation

Data presented in this paper were mainly derived from measurements with miniature five-hole pressure probes of overall diameter 1.7 mm, hole bore 0.25 mm, which were used in a nonnulled mode as outlined by Wray et al. [11]. Measurements were made 2 mm downstream of the OGV trailing edge, 1 mm downstream of prediffuser exit, and at up to three planes within each feed annulus (Fig. 2). At all these stations area traverses were performed. For measurements within the prediffuser and outer annulus, radial movement of the instrumentation was obtained by attaching the probes to stepper motors located on the outer casing. In addition, to avoid numerous circumferential instrumentation locations, the OGV row and flame tube could be indexed around to provide the required circumferential movement. For measurements within the inner annulus a traversing mechanism was mounted inside the flame tube. As described by Wray et al. [11] this could be rotated circumferentially, independent of the flame tube, while also providing radial movement of the probe. Using these techniques, data were recorded at 20 radial positions, and this was repeated at 21 circumferential locations for each traverse plane. The area traversed either corresponded to one OGV blade space, as at prediffuser inlet, or one burner sector for measurements within the feed annuli. At prediffuser exit measurements were made across an OGV space and a burner sector in order to define accurately the blade wakes and assess the influence of cowl head geometry at this location. Numerous static tappings were also located within the dump cavity, around the flame tube head, and on the casing walls within the feed annuli and turbine bleed passage. Stagnation pressure levels within the turbine bleed passage were derived from fixed pitot rakes. Operation of the test rig, positioning of the instrumentation, and digitizing of all pressure transducer signals was controlled by a Personal Computer (PC).

Data Reduction

At each traverse point, five-hole probes provide information on stagnation and static pressure in addition to velocity magnitude and direction. From this information the overall mean velocity at a traverse plane was calculated using an area-weighted method so that:

$$\bar{U} = \frac{1}{A}\int UdA \tag{1}$$

with the mass flow rate through a traverse plane being defined as

$$m = \int \rho UdA = \rho\bar{U}A \tag{2}$$

Spatially averaged values of stagnation and static pressures at a traverse plane were defined by mass weighting the appropriate individual values, i.e.,

$$\tilde{P} = \frac{1}{m}\int Pdm \quad \tilde{p} = \frac{1}{m}\int pdm \tag{3}$$

For radial distributions of circumferentially averaged quantities, this weighting was performed in the circumferential direction only. All flow angles presented were derived from the mass-weighted tangential and area-weighted axial velocity components as outlined by Dring and Spear [12]. It should also be noted that all of these averaged quantities were derived by numerical integration of the data to which had been fitted a spline curve.

Changes in the spatially averaged pressures between various planes within the diffuser system are expressed in terms of the stagnation pressure loss (λ) and static pressure recovery (C_p) coefficients:

$$\lambda_{a-b} = \frac{\tilde{P}_a - \tilde{P}_b}{\tilde{P}_a - \tilde{p}_a} \quad C_p = \frac{\tilde{p}_b - \tilde{p}_a}{\tilde{P}_a - \tilde{p}_a} \tag{4}$$

where "a" and "b" are the upstream and downstream planes, respectively.

Estimate of Experimental Errors

At a given data point the experimental accuracy is influenced by:

(i) the spatial error associated with the finite distance between the five holes on the probe tip; this was eliminated by radial and circumferential interpolation of the side pressures onto the central measurement hole [11].
(ii) the proximity of casing surfaces to a measurement location.
(iii) recording of the pressure signals via transducers and an analogue to digital convertor.
(iv) maintaining the compressor on its correct operating point.

Discrepancies in mass flow of less than 2 percent were recorded between the pre-diffuser inlet and exit planes due to these sources of error. Based on these values and the repeatability of test data it was estimated that the stagnation pressure loss (λ) and static pressure recovery (C_p) coefficients were accurate to within 0.01 of their true values. Flow angles were thought to be in error by up to 1 deg.

Results and Discussion

Results are presented for the diffuser in its datum configuration, with a dump gap of 1.0, and at three additional dump gaps of 0.80, 0.65, and 0.50.

Inlet Conditions. Axial velocity contours indicate the circumferential and radial distribution of fluid at inlet to the diffuser system (Fig. 3). As well as boundary layers on each casing, well-defined blade wakes are present from the upstream compressor outlet guide vanes. Within experimental error, these inlet conditions were the same for all the geometries tested.

Several authors have noted how the performance of a diffuser system varies with inlet conditions. For example, Stevens et al. [8] have shown that blade wakes can have a favorable effect by re-energizing boundary layer flow due to enhanced mixing.

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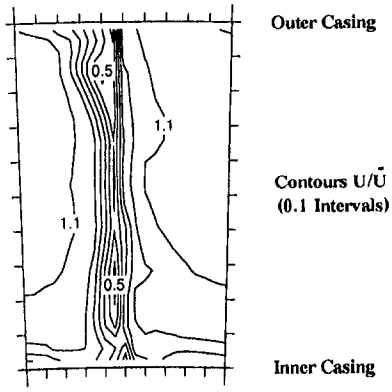

Fig. 3 Axial velocity distribution at prediffuser inlet

However, the presence and mixing out of such flow nonuniformities will lead to an increase in the diffuser system's mass-weighted pressure loss. To try and reduce this loss variation with inlet conditions Klein [9] defines diffuser performance in terms of momentum mix weighted values. In this case, overall stagnation and static pressures at the inlet and exit planes are those which would occur if the respective velocity profiles were to mix out in a frictionless duct of constant area. However, within a diffuser system the flow area increases and, as outlined by Denton [13], the resulting positive pressure gradient increases the mixing losses, which are not accounted for in the momentum mix approach.

It should be noted that all performance data presented in this paper are in terms of mass-weighted pressures and therefore include not only the favorable effects of wake mixing, on boundary layer flow, but also any mixing loss associated with the nonuniform inlet flow.

Datum Configuration. The stagnation pressure losses (λ) between prediffuser inlet ($X1$) and the outer ($XO3$) and inner ($XI3$) feed annuli are 0.483 and 0.430, respectively. Since the loss within the prediffuser is 0.068, this implies that a large proportion of the loss occurs downstream of the prediffuser as the flow passes around the flame tube head and enters the feed annuli. It is also interesting to note the relatively high stagnation pressure loss (λ), of 0.664, that is associated with the flow entering the turbine bleed passage. This represents low-energy high-loss flow that is drawn through the bleed ports from the inner dump cavity. The remaining flow passes to the inner annulus and so, by removing low-energy air, this may account for the relatively low loss within this annulus relative to the outer feed annulus. With this in mind it is interesting to note the combined mass weighted inner annulus and turbine bleed loss of 0.485 is virtually the same as the outer feed annulus loss of 0.483.

The prediffuser static pressure recovery coefficient (C_p) of 0.373 is a significant proportion of the total static pressure recovery values of 0.448 and 0.510 recorded between prediffuser inlet and the outer ($XO3$) and inner ($XI3$) feed annuli. Although there is a relatively large velocity reduction between prediffuser exit ($\approx$33 m/s) and the feed annuli ($\approx$11 m/s), the static pressure rise is significantly reduced by the relatively high stagnation pressure loss around the flame tube head. It can therefore be seen that in terms of overall diffuser performance, the majority of the static pressure rise occurs within the prediffuser.

It should be noted that diffuser performance is based on the overall mean flow properties at each traverse plane. However, there will be slight variations between these values and the actual loss and diffusion incurred along each flow path. This is because the actual values depend on which portion of the prediffuser flow passes into the feed annuli or turbine bleed passage. For example, assuming it is the 24.3 percent of the flow closest to the prediffuser outer casing that passes to the outer feed annuli, then a dividing streamline can be obtained and mass-weighted pressures for that streamtube calculated. This yields, for the flow passing to the outer annulus, a loss of 0.082 within the prediffuser (λout_{1-2}) and a total loss of 0.459 (λout_{1-03}). Applying a similar technique to the inner annulus flow gives, for the prediffuser and total loss, values of 0.048 (λin_{1-2}) and 0.408 (λin_{1-I3}), respectively. Comparison with the data based on overall mean flow properties indicates some differences, but variations are relatively small and show the same trend of high losses around the flame tube head. Similarity of the results is thought to reflect the lack of large radial flow distortions at inlet to, and within, the prediffuser as well as a comparable stagnation pressure loss at different radii. In addition, this streamtube method was first used by Fishenden and Stevens [1] whose flame tube incorporated no porosity. However, the head porosity in this investigation means that, as well as radial location, the flow path at prediffuser exit is also determined by whether it is in line with, or between, cowl holes. Thus, flow entering each feed annuli is more uniformly distributed at prediffuser exit and is not confined to streamtubes adjacent to each casing.

Flow Field Characteristics. Comparison of the axial velocity contours at prediffuser inlet (Fig. 3) and exit (Fig. 4) indicates decay of the compressor OGV blade wakes within the prediffuser. As discussed by Hill et al. [14], in large positive gradients the pressure forces can increase the velocity gradients since the momentum of the slower moving fluid undergoes a greater reduction compared with the faster moving fluid. However, decay of the blade wakes means that in this geometry it is the flow shear forces, rather than the pressure forces, which are predominant. The radial distribution of flow at inlet ($X1$) and exit ($X2$) from the prediffuser is more clearly indicated by the circumferentially averaged axial velocity profiles (Fig. 5). These are nondimensionalized by the calculated mean velocity at each plane and comparison of the profiles indicates the boundary layer growth that occurs along each prediffuser casing. This leads to a radial redistribution of fluid so that, at prediffuser exit, there is a relative increase in the amount of flow in the center of the prediffuser (between 10 and 90 percent of annulus height). The boundary layer growth is promoted by the positive pressure gradient within the prediffuser due to the increasing flow area. This pressure rise is shown by the static pressure profiles at each plane, which are nondimensionalized by the conditions at prediffuser inlet (Fig. 5). It should also be noted

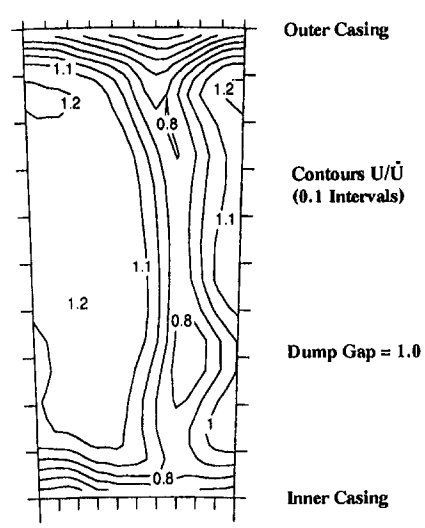

Fig. 4 Axial velocity distribution at prediffuser exit

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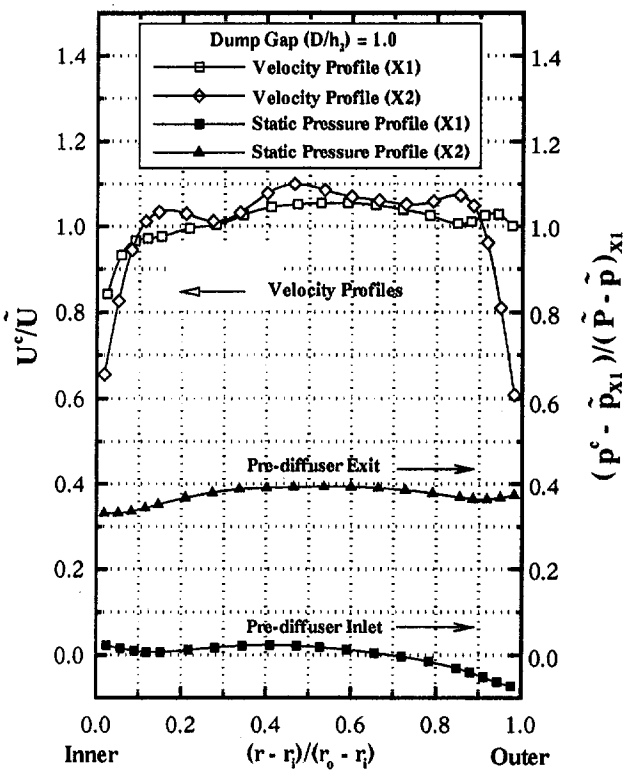

Fig. 5 Axial velocity and static pressure profiles

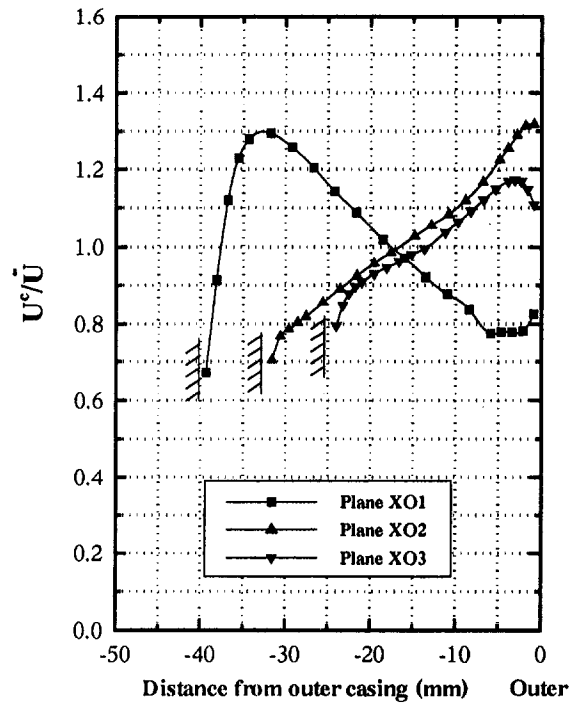

Fig. 7 Outer annulus axial velocity profiles (dump gap = 1.0)

that due to gradients of pressure in the radial direction, the pressure increase along each casing is different. This is thought to contribute to the more pronounced growth of the boundary layer along the outer casing.

At prediffuser exit the flow separates to form a recirculation zone in the dump cavity. This region bounds the mainstream flow, which is discharged into the dump region and impinges on the flame tube head. The flow accelerates around the head, away from the stagnation point, giving rise to high velocities adjacent to the cowl surface (Fig. 6). This is reflected by the measured static pressure distribution on this surface that is presented later (Fig. 12). The maximum velocity acquired is determined mainly by the curvature undertaken by the flow as it passes around the head and into the feed annuli. Measurements by Carrotte et al. [6] and Stevens et al. [2] have also shown how the velocity away from the cowl, toward the recirculation region, decreases rapidly. It is the high-velocity gradients in this region where a significant amount of turbulence, and hence stagnation pressure loss, is generated [6]. At entry to each feed annulus the flow distribution is still biased toward the flame tube, as shown for the outer feed annuli by the circumferentially averaged axial velocity profile at plane $XO1$ (Fig. 7). However, due to the unstable nature of the flow passing around the head, this bias rapidly changes toward the outer casing, as measured at plane $XO2$ downstream of the first cooling ring (Fig. 7). The turbulence in this region is relatively high and, in

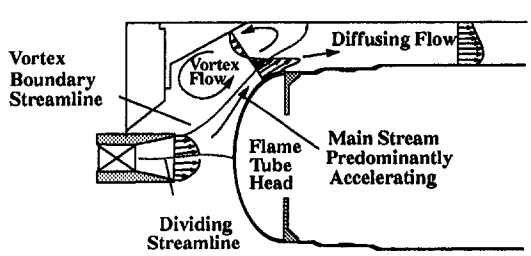

Fig. 6 Flow field downstream of pre-diffuser

several other investigations, has led to rapid mixing out of the velocity profile. Comparison of the profiles at stations $XO2$ and $XO3$ shows this is not apparent in the present investigation. However, it is thought the passage of air into the flame tube through primary ports and the secondary cooling ring helps maintain the distorted profile down the feed annulus, features that are often not simulated in other investigations. Although not presented, similar bias characteristics and axial velocity profiles have been measured for the flow passing down the inner annulus.

As well as the radial distribution of flow, significant variations in the swirl angle were found as the flow passed through the diffuser system (Fig. 8). At prediffuser inlet the increased swirl angles close to each casing (within 10 percent of the passage height) reflect the presence of flow overturning due to classical secondary flow within the upstream OGV passage. This is a usual occurrence within the stationary blade rows of axial compressors. Flow overturning by the OGVs, of approximately 3 deg, is also indicated within the central region of the passage. The presence of this tangential momentum at inlet to the system leads to significant changes in the swirl angles (θ) farther downstream, particularly for the flow passing to the inner annulus (Fig. 8). This is due to the change in radius, relative to the engine centerline, that is undertaken by the flow within the diffuser system and the reduction in axial velocity. For example, if tangential momentum is conserved, then the product of a particle's tangential velocity and radius ($U_t r$) is constant so that, as $\tan\theta = U_t/U$, then $\tan\theta \propto 1/Ur$. Within the prediffuser, the mean radius is constant and so the increase of swirl angle at prediffuser exit is mainly due to the reduction in axial velocity (U). For flow passing to the outer annulus, a further reduction in axial velocity occurs, but this is also accompanied by an increase in radius (r) leading to minimal change in swirl angle. In contrast, flow entering the inner feed annulus undergoes both a reduction in axial velocity and radius so that flow swirl significantly increases. This effect may be enhanced further if the flow close to the inner casing of the prediffuser, which has been overturned within the OGV passage, passes into the inner annulus. The calculated tangential momentum per unit mass flow at prediffuser inlet (0.85 $m^2 s^{-1}$), exit (0.83 $m^2 s^{-1}$), and within the outer (0.36 $m^2 s^{-1}$) and inner (0.49 $m^2 s^{-1}$) feed

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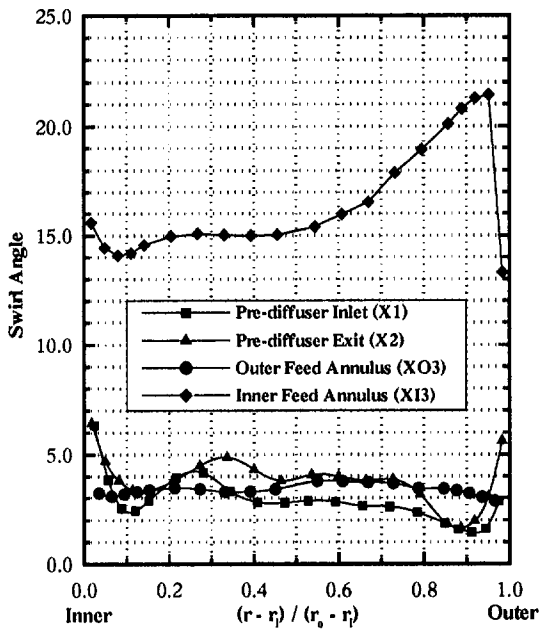

Fig. 8 Swirl angle profiles through the diffuser (dump gap = 1.0)

annuli was calculated using a similar method to that outlined by Dring [15]. Due to friction it can be seen that a relative decrease of tangential momentum occurs through the diffuser system. Although this reduces the magnitude of the swirl, angles in excess of 15 deg are still recorded within the inner annulus. The presence of such swirl is thought to have only a minor effect on the diffuser's static pressure rise and stagnation pressure loss. However, of greater significance is thought to be its effect on the discharge coefficients for the inner annulus cooling rings, primary and dilution ports as well as the behavior and stability of the flow passing through these features. Carrotte and Stevens [16] have already indicated how vortices and other flow instabilities can develop within dilution holes and affect mixing within the combustion chamber. It should also be noted that as radial flame tube depth increases, so this amplification of swirl angle will become greater. Furthermore, for a given geometry the magnitude of the angles within the inner annulus is determined by the swirl at diffuser inlet, and in a gas turbine engine this will vary with engine speed.

Effects of Dump Gap Variation. The variation of stagnation pressure loss to the turbine bleed passage (Xb), outer ($XO3$), and inner ($XI3$) feed annuli increases significantly as the dump gap is reduced (Fig. 9*a*). For example, at a dump gap of 0.50 the loss to the inner and outer feed annuli increases by 48 and 32 percent, respectively, relative to the datum configuration. Since the overall performance of the prediffuser does not vary significantly and the loss within each feed annulus is small, virtually all the increase in stagnation pressure loss occurs within the vicinity of the flame tube head (i.e., between stations $X2$ and $XO1$, $XI1$). In addition to the increased stagnation pressure loss, there is also a corresponding decrease in static pressure recovery (Fig. 9*b*). Although a small proportion of this reduction is associated with more distorted velocity profiles within each feed annulus, the majority of this effect is due to the higher stagnation loss. For all the geometries tested, the static pressure rise within the prediffuser is virtually the same, while at large dump gaps there is an additional pressure rise between prediffuser exit ($X2$) and the feed annuli. As the dump gap reduces, the increased stagnation pressure loss around the head reduces the static pressure rise. Eventually, at dump gaps less than approximately 0.65, the loss is of sufficient magnitude that the static pressure falls downstream of prediffuser exit.

Prediffuser Performance. The overall prediffuser performance does not vary significantly with dump gap, but changes to the flow field do occur in this region. At prediffuser exit the circumferentially averaged radial static pressure profiles, nondimensionalized by the mean flow properties at the inlet plane, indicate that reducing the dump gap causes an increase of static pressure in the central region of the passage and a decrease toward each casing (Fig. 10). This reflects the pressure distribution that is generated by flow curvature around the flame tube head, which, due to the flame tube's upstream influence, becomes apparent at prediffuser exit when the dump gap is small. Fishenden and Stevens [1] also noted this effect and, since inlet conditions are not affected, means that diffusion is increased along the center streamtube of the prediffuser but decreased close to each casing. These effects are also reflected by the axial velocity profiles at prediffuser exit (Fig. 11), the midpassage velocities decreasing with a corresponding rise in velocity close to each casing.

The lower static pressure gradient along the prediffuser casings at small dump gaps reduces boundary layer growth. Hence, as dump gap is reduced, it should be possible to introduce corresponding increases in prediffuser area ratio, within the same axial length, and not adversely affect the condition of the boundary layers on each casing. This should increase the static pressure recovery within the prediffuser and help offset the larger stagnation pressure losses around the flame tube head. With this in mind it is also interesting to note the work of Hestermann et al. [5], who unstalled a prediffuser by operating the system at a reduced dump gap.

Dump Cavity Flow. It is in the region surrounding the flame tube head where most of the loss increase occurs as the dump gap is reduced. Although instrumentation is limited, cowl sur-

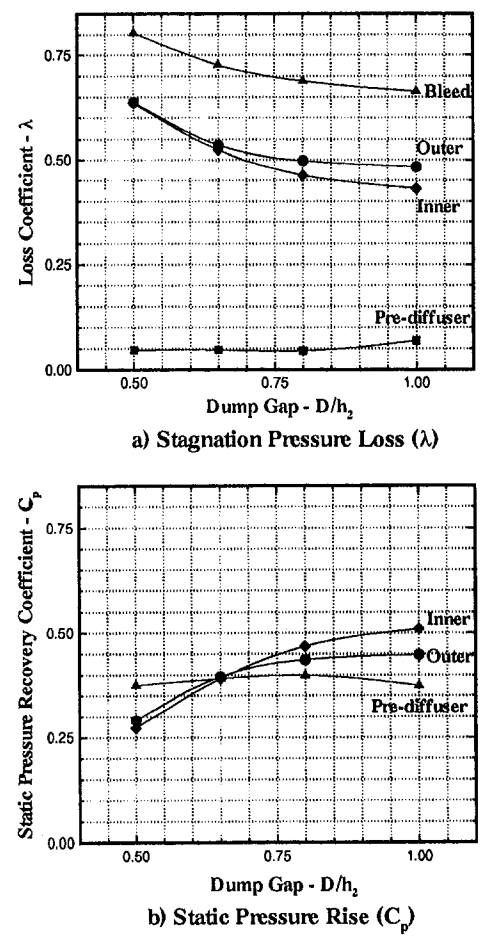

Fig. 9 Effect of dump gap on diffuser performance

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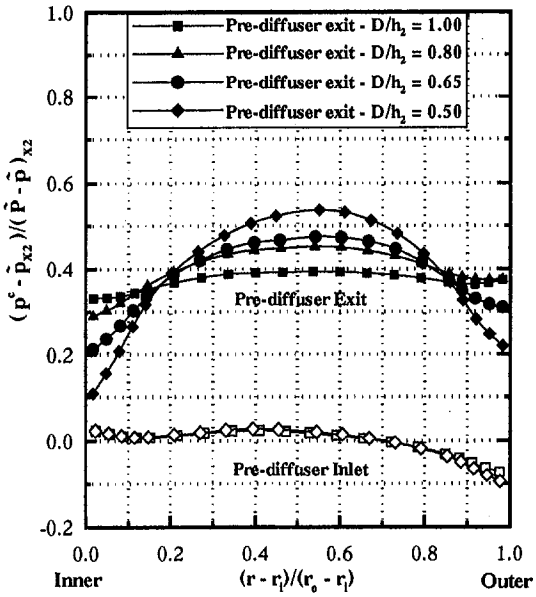

Fig. 10 Prediffuser exit static pressure profiles

face tappings allow the static pressure distribution, averaged over one OGV blade space, to be determined around the head. It can be seen that away from the stagnation point, the pressure decreases rapidly to a minimum before undergoing a rise in pressure prior to entering each feed annulus (Fig. 12). This static pressure distribution reflects the velocities adjacent to the cowl surface, which accelerate from zero at the stagnation point up to a maximum value, at the minimum pressure location, prior to diffusing into the downstream annuli. In addition, as measured by Carrotte et al. [6] away from the flame tube head, toward the recirculation region, the velocity decreases rapidly giving rise to steep velocity gradients. It is these gradients where turbulent energy is produced and the stagnation pressure loss generated. As the dump gap is reduced the minimum pressure on the head decreases, indicating higher velocities adjacent to the surface. A corresponding increase of the velocity gradients around the head must therefore occur, thus explaining the increase in stagnation pressure loss. The higher velocities and

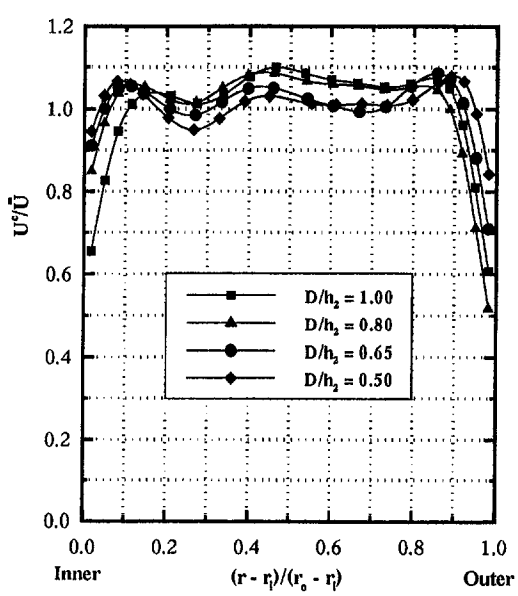

Fig. 11 Prediffuser exit axial velocity profiles

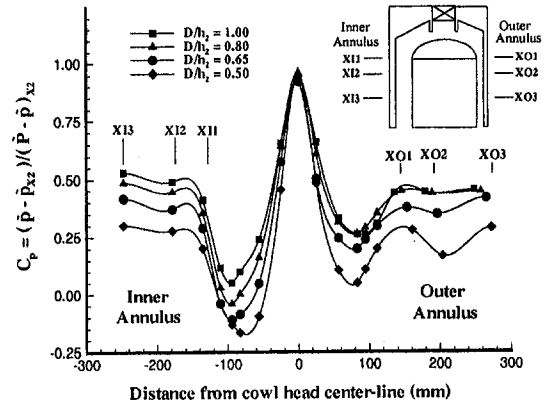

Fig. 12 Flame tube head static pressure distribution

steeper velocity gradients are due to the greater curvature and more rapid rates of deflection that must be undertaken by the flow, at small dump gaps, in order to pass around the flame tube head and into the feed annuli. These more distorted profiles are also indicated by the axial velocity profiles within the downstream annuli (Fig. 13). It should also be noted that as dump gap is reduced, the stagnation pressure loss incurred by fluid passing to the inner annulus increases at a greater rate compared with the outer annulus flow. This is also reflected by the greater reduction in minimum static pressure on the inner annulus side of the cowl surface (Fig. 12). One possible reason for this effect is that flow passing to the outer annulus is subjected to an increasing radius and can therefore diffuse circumferentially around the head. This is in contrast to flow passing to the inner annulus where the decreasing radius may lead to higher cowl velocities, steeper velocity gradients, and increased losses. Furthermore, downstream of the minimum pressure location the rapid rise in pressure, at small dump gaps, means the flow in this region is prone to separate from the cowl surface.

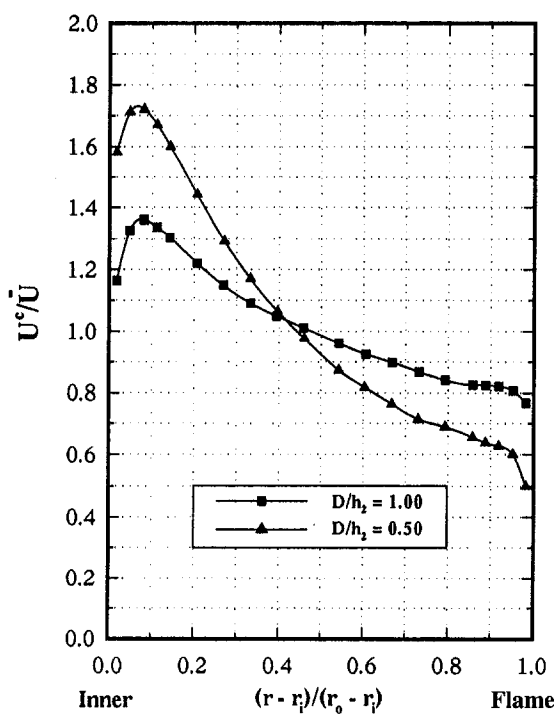

Fig. 13 Inner annulus axial velocity profiles (XI3)

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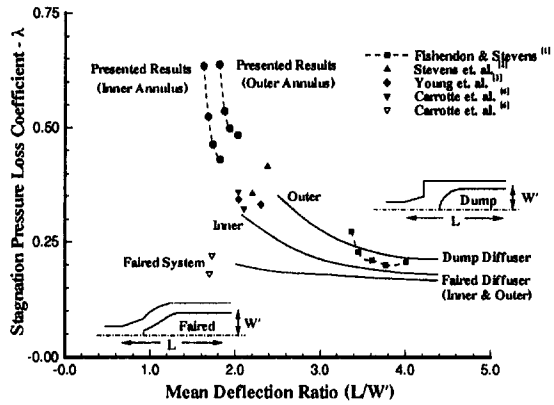

Fig. 14 Comparison of combustor diffuser system losses

Comparison With Other Data. The stagnation pressure losses from a variety of published data are presented (Fig. 14) relative to the ratio of diffuser length (L) to flow deflection (W'). Diffuser length is defined as the axial distance between prediffuser inlet and the combustor primary ports, whereas flow deflection is based on the radial distance between the diffuser centerline and the mean radial location of each feed annulus. This latter parameter is therefore mainly a function of the flame tube radial depth. The diagram contains a wide variety of results from various diffuser systems with differing inlet conditions, various amounts of diffusion, flame tubes with and without porosity, etc. In addition, the definition of diffuser length and deflection can be somewhat subjective, particularly for systems with large cant angles. However, a correlation is suggested between the results, although for a given geometry, minor loss variations will occur due to changes in mass flow split, prediffuser area ratio, cowl shape, etc. The general trend indicates that for diffusers with relatively long lengths and narrow flame tubes, the losses are relatively low and of comparable magnitude to those of a faired system [1]. However, the deeper flame tubes and shorter lengths inherent to more modern systems result in larger flow deflections and a more severe radius of curvature as the flow passes around the head. As already discussed and demonstrated by the results, this leads to increased pressure losses. For diffusers typical of the geometry tested in this paper, any further reductions in length or increases in flame tube depth will lead to relatively large pressure loss penalties. Hence, any potential benefits in terms of weight reduction, for example, may be more than offset by the adverse effect on engine specific fuel consumption.

Included in the comparison are the results of Carrotte et al. [6] obtained for a short faired system. This indicates that for relatively short diffusers with deep flame tubes, there are large differences in loss performance between dump and faired systems. Due to their stability characteristics, it is not thought practical or desirable to operate such faired systems within a gas turbine engine. However, as diffuser systems become shorter and flame tubes deeper, it may be desirable to investigate a possible hybrid or compromise system. This would utilize some of the potential loss reduction offered by a faired system while still achieving acceptable stability performance.

Conclusions

An experimental investigation has been undertaken on a modern dump diffuser system with inlet conditions being generated by an axial flow compressor. Measurements were obtained at a dump gap of 1.0 (datum configuration) and at three other dump gaps (0.80, 0.65, and 0.50). The following conclusions have been drawn:

- For the datum configuration, the stagnation pressure loss coefficients (λ) between prediffuser inlet and the outer and inner feed annuli were 0.483 and 0.430, respectively. The majority of this loss was generated around the flame tube head while most of the static pressure rise occurred within the prediffuser.
- The stagnation pressure losses increased by 32 (outer) and 48 percent (inner) as the dump gap was reduced from 1.0 (datum) to 0.50.
- The high losses were associated with the relatively deep flame tube and short system length. This increases the amount of flow curvature around the flame tube head, which becomes more severe as dump gap is reduced.
- For all the geometries tested, relatively high swirl angles ($\approx$15 deg) were observed within the inner feed annulus compared with those at prediffuser inlet ($\approx$3 deg). This amplification is due to the decrease in axial velocity and radius that is undertaken by the flow passing to the inner annulus.

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